



NovoRetin™

The natural alternative to retinol

Retinol – the Almost Perfect Anti-Aging Compound



- Huge trend for retinol (vitamin A) and its derivatives in cosmetics.
- Retinol and structurally related molecules of the group of retinoids are important for many **biological processes**, including cell growth and cellular differentiation, and thus wound healing, barrier function and the maintenance of the skin in general.
- They naturally occur in human organs including the skin.
- Topically applied, they provide strong **anti-aging** properties and are used to treat skin disorders such as **acne**, **psoriasis** and **ichthyosis**.
- Most used and most studied anti-aging compounds!

Retinol – the Almost Perfect Anti-Aging Compound



- Retinol can cause **skin dryness** and **irritations** and increases the skin's **sensitivity** to sunlight.
 - Retinol is **not stable** in cosmetic formulations.
 - Derivatives (retinaldehyde, retinyl palmitate) are more stable, but perform much worse in terms of activity.
 - Retinoic acid, the most important active metabolite of retinol, is **highly irritative** → not allowed to use in cosmetics.
- **Plant-based, more gentle alternatives** are on high demand on the market.
- Mibelle Biochemistry developed a new active with **retinol-like activity** based on mastic.

Mastic from the Mastic Trees of Chios



- Chios mastiha also called **mastic** is an aromatic resin produced by the lentisk tree that grows on the Greek island of Chios (*Pistacia lentiscus* var. Chia).
- Evergreen shrub which grows on arid, rocky and poor soil.
- Only the mastic trees of southern Chios give the **mastiha resin** (soil and especially warm and dry weather conditions).
- The know-how to cultivate mastic in Chios is considered as “**intangible cultural heritage of humanity**” by UNESCO in 2014 and its production is granted “**protected designation of origin**” (PDO).

Mastic, the Gold of Chios



- Mastic is excreted in the form of **tears** from the trunk and main branches when they are gently scratched on the surface.
- The extraction of the resin through this technique is a **sustainable** and **centuries-old tradition** that does not harm the tree and can be practiced for decades.
- Solidification of the resin results in an ivory translucent colored drop (tear), which is collected, washed and cleaned.
- The name “mastic” is derived from the Greek words “μαστάζω” = *chew* and “μάσταξ” = *mouth*.

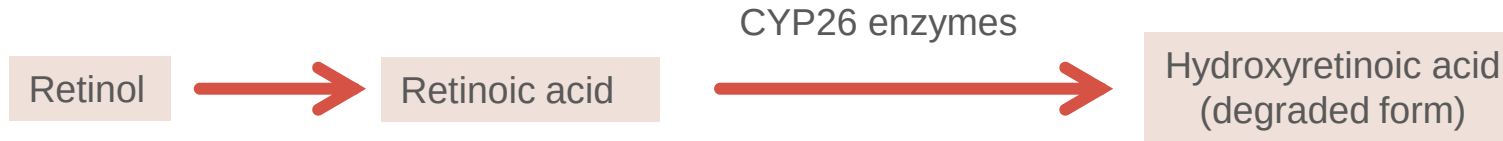
Mastic is the Chewing Gum of the Ancient World



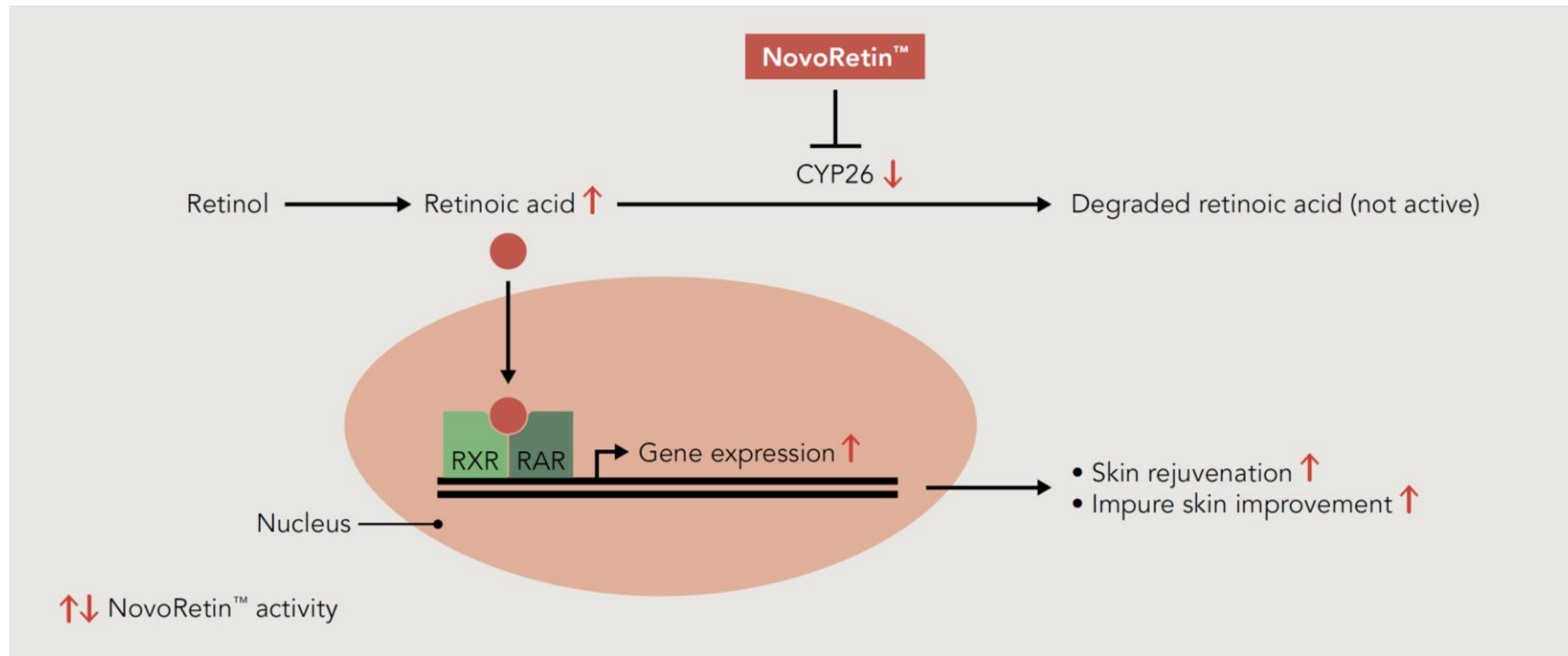
- Mastic has been recognized since antiquity for its unique aroma and **therapeutic** properties.
- Mastic was the first **natural chewing gum** in the ancient world, used to clean the teeth and freshen the breath but also to prevent digestive disorders.
- Due to its high price, mastic was only bought by wealthy people.

Retinoic Acid Boosting Effect

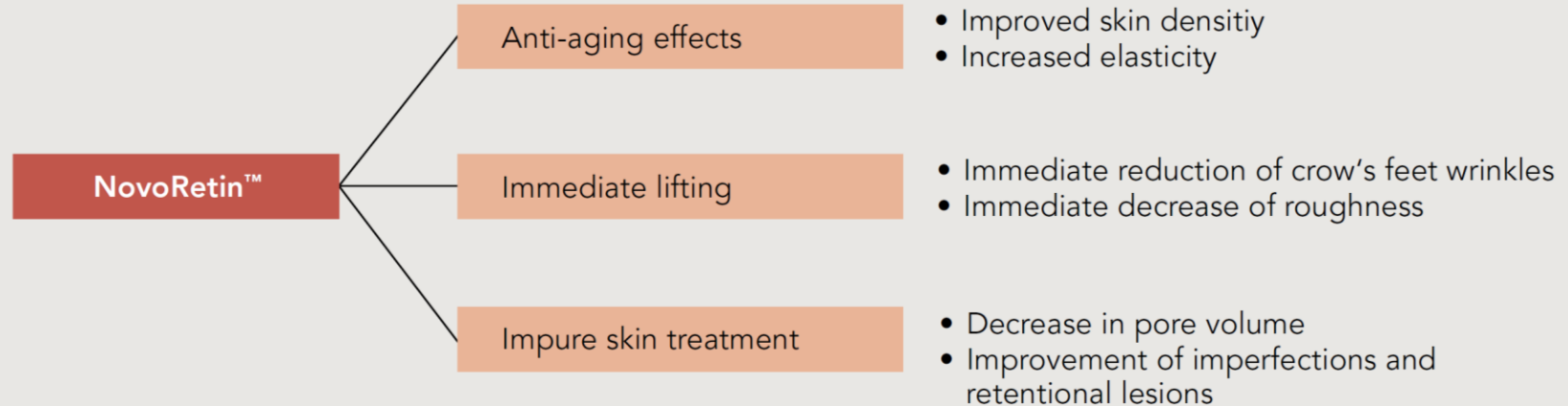
- The new active is based on an endogenous process that is present in every cell in our body: Vitamin A (retinol) is ingested through food → break down to retinoic acid if required.
 - In the cells, retinoic acid induces its own degradation by activating an enzyme of the CYP26 family.
 - In **skin** the enzyme **CYP26A1** is responsible for the elimination of retinoic acid.
- By **inhibiting CYP26A1 enzyme**, it is possible to **increase the endogenous levels of retinoic acid in the skin**.



Retinoic Acid Boosting Effect of NovoRetin™



Activities of NovoRetin™



NovoRetin™ Composition

Chios Mastic	2.3 %
Lecithin	2 %
Pentylene Glycol	2.7 %
Glyceryl Caprylate	0.3 %
Caprylic/ Capric Triglyceride	12.7 %
Deionized Water	ad 100 %

INCI / CTFA-Declaration

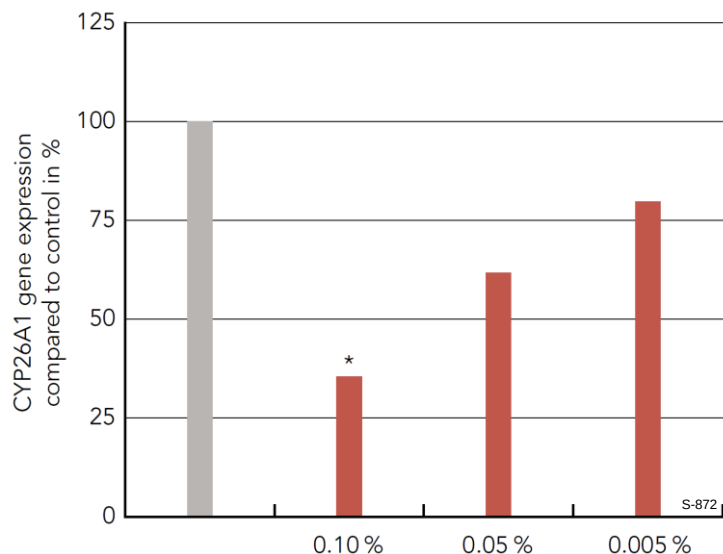
Pistacia Lentiscus Gum / Pistacia Lentiscus (Mastic) Gum (and) Lecithin (and) Pentylene Glycol (and) Glyceryl Caprylate / Caprate (and) Caprylic / Capric Triglyceride (and) Aqua/Water

Recommended Use Level: 2 – 3 %



Inhibition of CYP26A1 Gene Expression

■ Control ■ NovoRetin™



*p<0.05 versus control

Cell culture: Human keratinocyte cell line (HaCaT)
Test substance: NovoRetin™, different concentrations
Treatment: Treatment of cell cultures with test substance for 4 hours
Parameter: Gene expression of CYP26A1 (RT-qPCR)

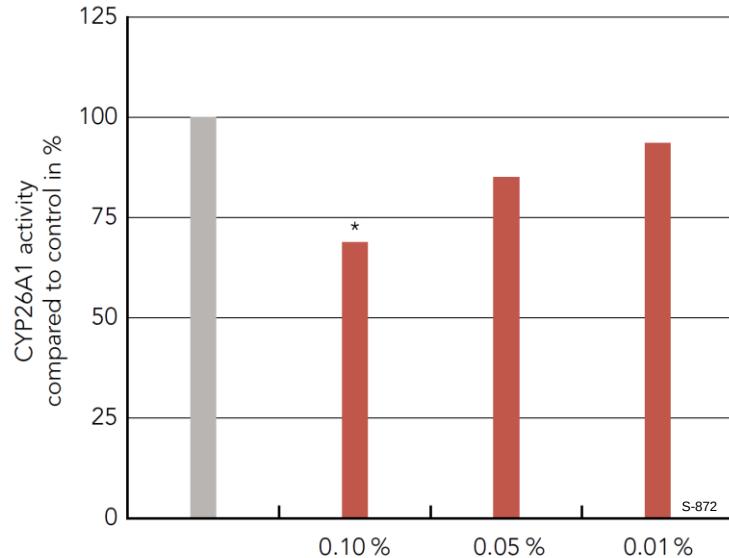


→ Clear concentration-dependent reduction in the expression of CYP26A1.



Inhibition of CYP26A1 Activity

■ Control ■ NovoRetin™



*p<0.01 versus control

Cell culture: Human keratinocyte cell line (HaCaT)
Test substance: NovoRetin™, different concentrations
Treatment: Treatment of cell cultures with the test substance for 9 hours
Parameter: CYP26A1 activity (radiolabeling)



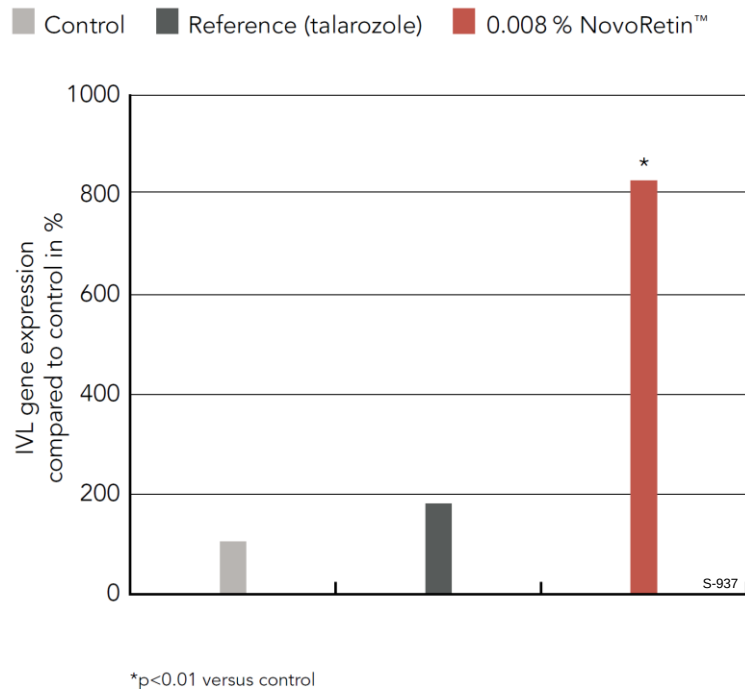
- Concentration-dependent reduction in the activity of CYP26A1.
- Both experiments indicate that NovoRetin™ can **increase the amount of endogenous retinoic acid** in the skin by delaying its rapid degradation → **retinol-like activity**.



Retinoic Acid Boosting Effect in 3D Epidermis Model

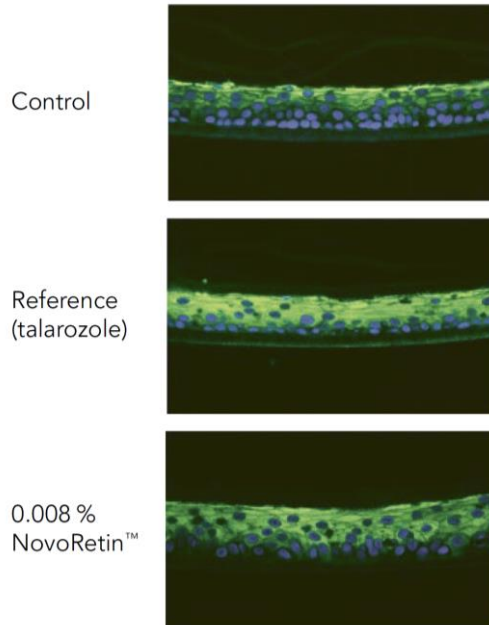
Skin model:	Reconstructed human epidermis (RHE)
Test substance:	0.008 % NovoRetin™
Treatment:	Treatment of skin models with test substance or the reference compound talarozole (10 nM), a well-known inhibitor of CYP26 enzymes
Parameter:	- Gene expression of involucrin, which is known to be upregulated by retinoic acid and marker for retinoic acid activity (RT-qPCR) - Protein abundance of involucrin in the RHEs (immunostaining and microscopy)

Retinoic Acid Boosting Effect in 3D Epidermis Model



Strong upregulation of IVL gene expression by +720 % compared to control.

Retinoic Acid Boosting Effect 3D Epidermis Model



● Involucrin protein ● Nuclear counter staining



- NovoRetin™ clearly **enhanced the abundance of involucrin** in the epidermis compared to control.
- NovoRetin™ has a **retinoic acid boosting effect** that is similar to the effect of the CYP26 inhibitor talarozole.

→ Topical application of NovoRetin™ can enhance retinoic acid levels in the skin.

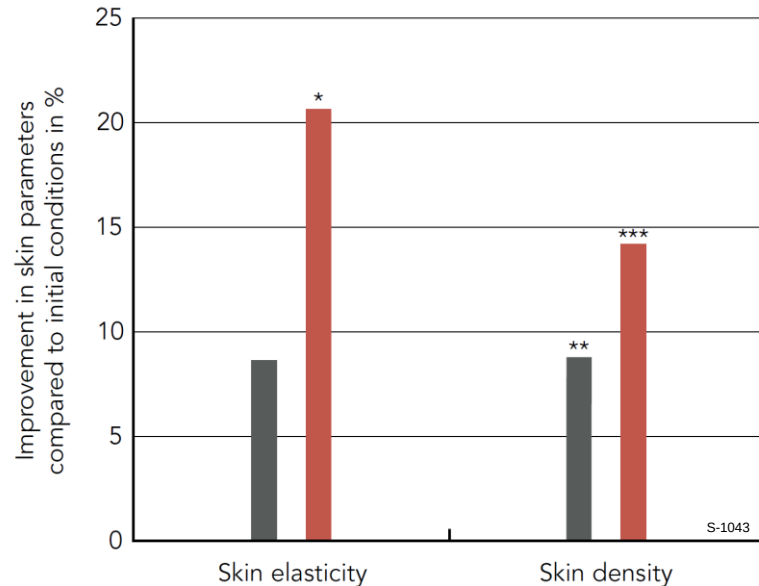


Anti-Aging Effect with NovoRetin™

- Test product:** Cream with 2 % NovoRetin™ or placebo
- Volunteers:** 18 (f, 40 y – 70 y) with wrinkles on the crow's feet region and signs of photo-aging on the face and inner forearms
- Application:** Application on the face and inner forearms twice daily for 28 days
- Parameters:**
- Skin elasticity in the face (Cutometer)
 - Epidermis-dermis density on the forearm (Dermalab® Ultrasound)

Increased Skin Density & Skin Elasticity

■ Placebo ■ 2 % NovoRetin™



*p<0.001 versus initial conditions

**p<0.01 versus initial conditions

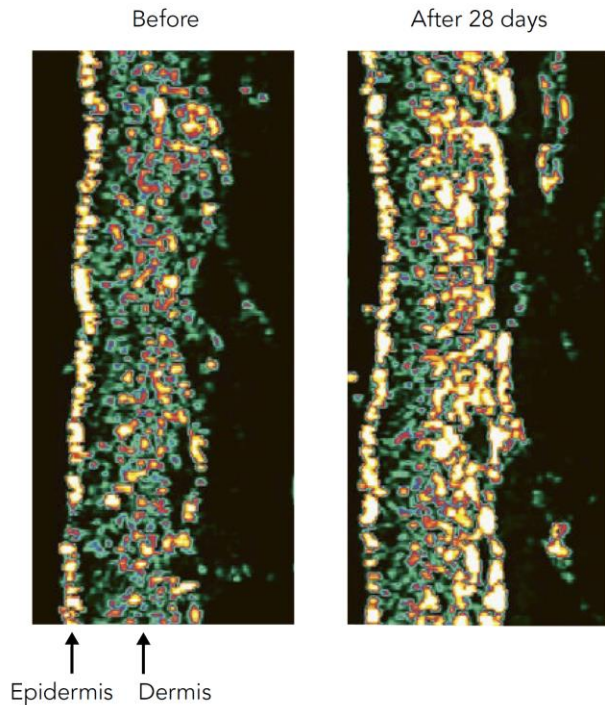
***p<0.001 versus initial conditions and p<0.05 versus placebo



NovoRetin™ significantly improved

- skin elasticity by +20.4 %
 - skin density by +13.8 %
- after 28 days of application.

Increased Skin Density



Improved epidermis-dermis density was also visible in the ultrasonographic images after 28 days of NovoRetin™ application.

→ By retaining endogenous retinoic acid levels in the skin, NovoRetin™ demonstrates strong anti-aging effects.

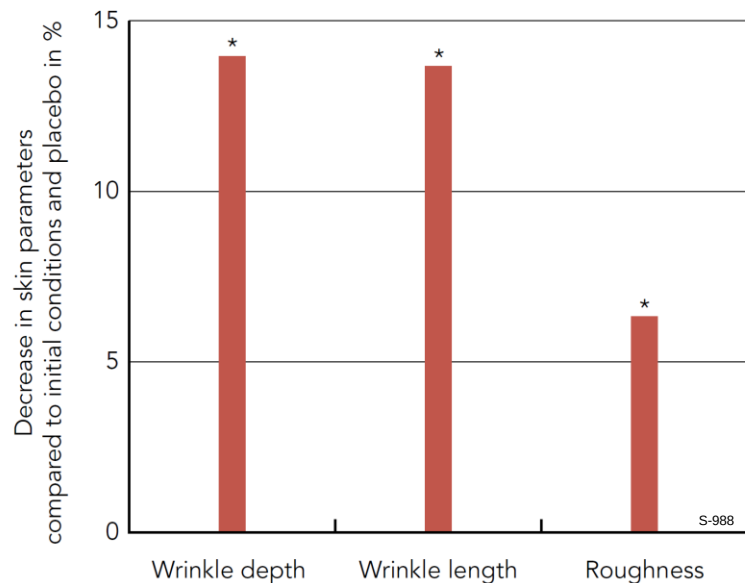


Immediate Lifting Effect with NovoRetin™

Test product:	Cream + 2 % NovoRetin™, placebo
Volunteers:	22 (f, 42 y – 65 y) with wrinkles in the crow's feet region
Application:	Single application on the periorbital area, measurements after 30 min
Parameters:	- Wrinkle depth, wrinkle length and skin roughness (Bio3D Structured-light Scanner, Bionos Biotech) - 2D images - Questionnaire

Immediate Anti-Wrinkle and Smoothing Effect

■ 2 % NovoRetin™



*p<0.05 versus initial conditions and placebo



NovoRetin™ was able to significantly improve:

- wrinkle depth by 14.0 %
- wrinkle length by 13.7 %
- and skin roughness by 6.4 %

after only a single application.

Visible Lifting Effect after a Single Application

Before

After 30 minutes



Photographs

Visible reduction of wrinkles in photographs taken of the volunteers before and 30 minutes after the application.

Questionnaire

82 % of the volunteers confirmed that the contour of their eyes appeared smoothed and that the eye region was immediately lifted.

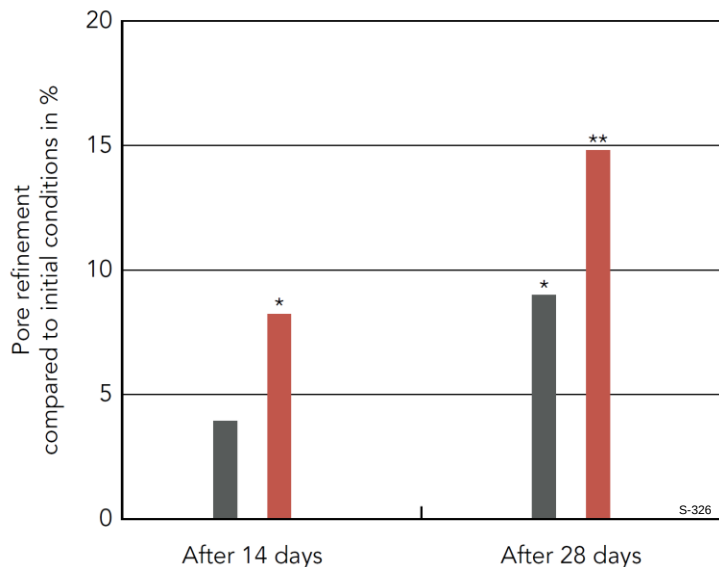


NovoRetin™ not only provides long-term anti-aging effects due to its retinoic acid-boosting efficacy but also offers **immediate lifting effects** in the crow's feet area after only a **single application**.



Pore Refinement

■ Placebo ■ 2 % NovoRetin™



*p<0.05 versus initial conditions

**p<0.05 versus initial conditions and placebo

Volunteers: 20 Caucasian women (24 to 61 y, mean: 42.8 y) with enlarged pores

Test product: 2 % NovoRetin™, placebo

Application: Twice daily for 28 days

Test area: Face

Parameter: Pore refinement (skin roughness, PRIMOS)

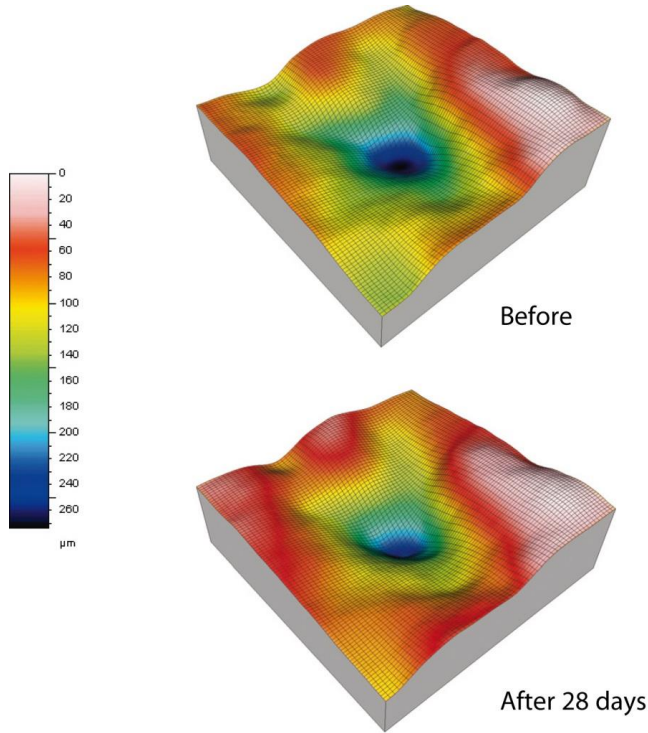


NovoRetin™ refined pores

- by 8 % after 14 days
- by almost 15 % after 28 days

Pore refining effect in 95 % of the volunteers.

Pore Size Reduction



Pore size: total pore area was determined by further calculations of the PRIMOS results.



Reduction of the total pore size in 75 % of the volunteers.

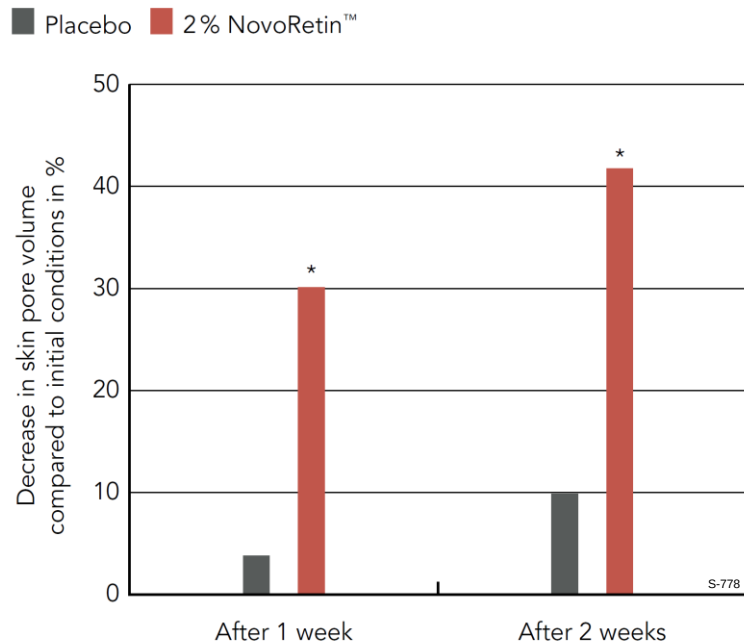
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NovoRetin™ Improves Pore Appearance

Test product:	Cream + 2 % NovoRetin™, placebo
Volunteers:	23 (f, 21 y – 48 y) Asian, with oily skin with impurities
Application:	Twice daily for 14 d, on one half of the face and the corresponding placebo on the other half of the face.
Parameters:	- Pore volume and pore count (cheeks, ANTERA 3D) - Pore appearance (cheeks, ANTERA 3D)

Improvement of Pore Volume



*p<0.05 versus initial conditions and placebo



Pore volume

2% NovoRetin™ significantly reduced pore volume by

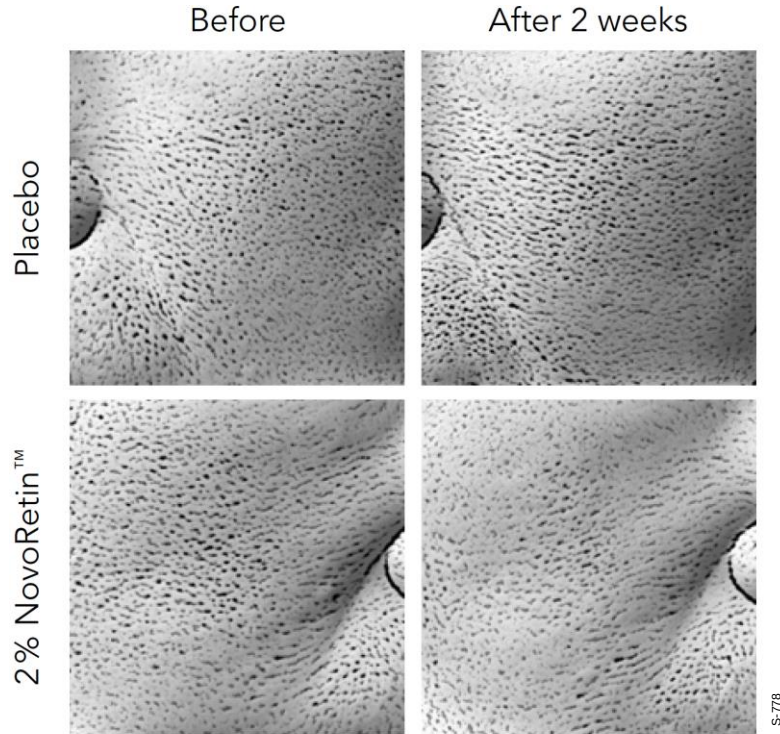
- 30.3 % after 1 week
- 41.7 % after 2 weeks

Pore count

Additionally, NovoRetin™ reduced pore count by

- 20.5 % after 1 week
- 30.3 % after 2 weeks

Visible Improvement of Pore Appearance



S-778



The NovoRetin™ treated side displayed a visible **improvement in pore appearance** after 2 weeks.

→ Effective reduction in the appearance of pores and an associated **improved skin texture** after just two weeks of using NovoRetin™.

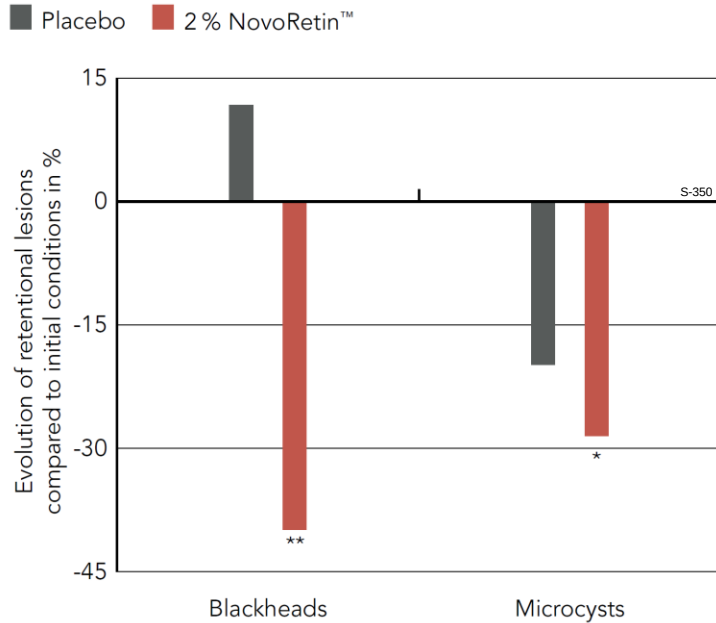
The placebo and NovoRetin™ pictures were taken from both halves of the face of the same volunteer.



Anti-Comedogenic Effect

Volunteers:	44 Thai women (30 y - 52 y) with retentional lesions (blackheads and microcysts) on their faces (blackheads and microcysts) on their faces, separated in two groups
Test product:	2 % NovoRetin™ emulsion, placebo
Application:	Twice daily for 28 days
Test area:	Face
Parameters:	- Comedones (blackheads) and microcysts (counted by dermatologists) - Shininess (scored visually by trained assessors) + macrophotographs

Anti-Comedogenic Effect of NovoRetin™



*p<0.001 versus initial conditions

**p<0.005 versus initial conditions and placebo



NovoRetin™ reduced the number of blackheads by 40 %, microcysts were reduced by 29 %.

Visible Reduction of Imperfections and Shininess



Before



After

→ Clarifying and matifying effect of NovoRetin™

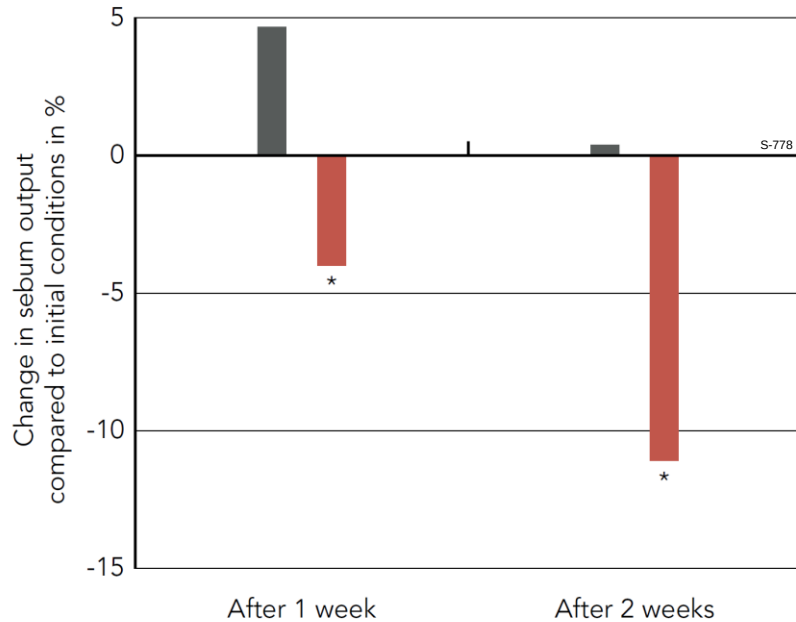


Improved Sebum Output and Skin Hydration with NovoRetin™

- Volunteers:** 23 (Asian, f, 21 y – 48 y) with wrinkles in the crow's feet region
- Test product:** Cream + 2 % NovoRetin™, placebo
- Application:** Application of the test product on one half of their face and the corresponding placebo on the other half of their face twice daily for 14 days.
- Parameters:**
- Sebum production (Sebumeter)
 - Skin hydration (Corneometer)

Reduced Sebum Production

■ Placebo ■ 2 % NovoRetin™



*p<0.05 versus placebo

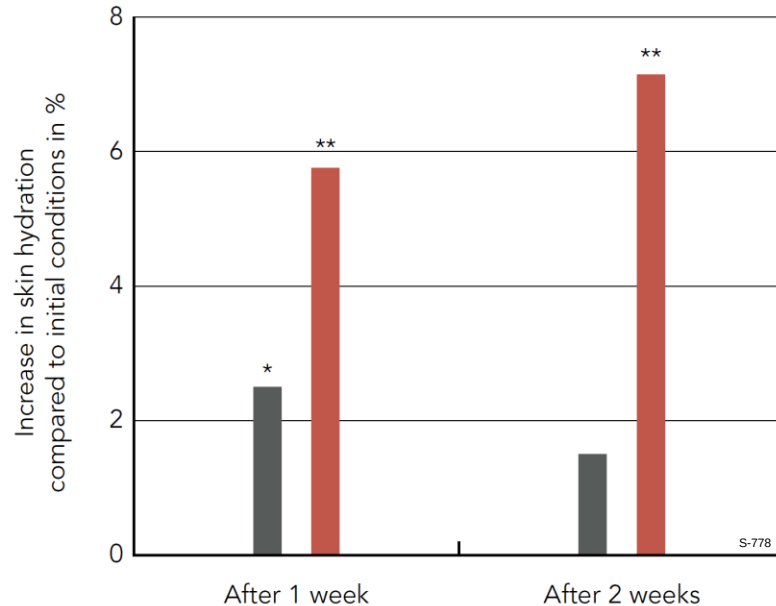


2 % NovoRetin™ reduced sebum output by

- 4 % after one week and
- 11 % after two weeks of application.

Increased Skin Hydration

■ Placebo ■ 2 % NovoRetin™



*p<0.05 versus initial conditions

**p<0.005 versus initial conditions and placebo



NovoRetin™ significantly increased skin hydration.

→ Despite efficiently reducing sebum production in oily skin types and in contrast to classic retinol applications, treatment with **NovoRetin™** **does not dry out the skin**. On the contrary, skin hydration is even promoted.

NovoRetin™



- Boosts naturally occurring retinoic acid in the skin
- Improves skin elasticity and density
- Effects immediate lifting
- Visibly refines pores and blemishes
- Stable and safe retinol activity

NovoRetin™ Applications



- Advanced age-defying formulas
- Immediate lifting products
- Treatments for acne-prone skin
- Pore minimizing formulas

NovoRetin™

Marketing Benefits



- A plant-based and gentle retinol alternative
- Very special and endemic source from the Mediterranean region
- All-in-one: increases skin density and elasticity, reduces pore size and wrinkles, refines skin texture, and creates a smooth matte finish
- Gentle yet highly effective – without the risk of increased photosensitivity and drying out the skin
- 100 % natural origin

NovoRetin™

Additional Marketing Material



Guide Formulation

Advanced Anti-Aging Sheet Mask with NovoRetin™
(only available on PDF)

PDFs

- Sourcing presentation

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